

Part 4: Springs in Series and Parallel

Configuration	Mass m (kg)	Frequency f (Hz)	$\omega = 2\pi f$ (rad/s)	$k_{\text{eff}} = m\omega^2$ (N/m)
Single spring	0.05	3.5385	22.23305121	24.7154283
Series	0.05	2.4615	15.46606063	11.95995158
Parallel	0.1	3.61538	22.7161025	51.60213126

Expected k_{series} (N/m)	12.35771415
Expected k_{parallel} (N/m)	49.43085661
Measured k_{series} (N/m)	11.95995158
Measured k_{parallel} (N/m)	51.60213126
% diff series	3.218739089
% diff parallel	4.39254911